

Problems on Frobenius Method

Textbook: Advanced Engineering Mathematics by Alan Jeffrey

1. Identify the nature of singularity of the following differential equations:

(i) $2x \frac{d^2 y}{dx^2} + (x+1) \frac{dy}{dx} + y = 0$ (Example 8.8)

(ii) $x^2 \frac{d^2 y}{dx^2} + x(2+x) \frac{dy}{dx} - 2y = 0$ (Example 8.9)

(iii) $x^2 \frac{d^2 y}{dx^2} + (x^2 - x) \frac{dy}{dx} + y = 0$ (Example 8.10)

(iv) $x^2 \frac{d^2 y}{dx^2} + (x^3 - x) \frac{dy}{dx} + y = 0$ (Example 8.11)

(v) $4x^2 \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} + (x-2)y = 0$ (Exercise 8.4)

$$c_1 = 1, c_2 = -\frac{1}{2}$$

(vi) $2x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - (2x+1)y = 0$ (Exercise 8.4)

$$c_1 = 1, c_2 = -\frac{1}{2}$$

(vii) $(2x + x^3) \frac{d^2 y}{dx^2} - \frac{dy}{dx} - 6xy = 0$

(viii) $x \frac{d^2 y}{dx^2} + (x-1) \frac{dy}{dx} - y = 0$

(ix) $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - 4)y = 0$

(x) $\frac{d^2 y}{dx^2} - x^2 \frac{dy}{dx} - y = 0$

(xi) $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - 1)y = 0$

(xii) $2x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + (1 - x^2)y = 0$

(xiii) $9x(1-x) \frac{d^2 y}{dx^2} - 12 \frac{dy}{dx} + 4y = 0$

(xiv) $3x \frac{d^2 y}{dx^2} + \frac{dy}{dx} - y = 0$

Hence solve the above differential equations in series by Frobenius method.

Legendre Polynomials

Adrien-Marie Legendre(1752-1833) was a distinguished French mathematician. His work on elliptic integrals provided basic analytic tools for mathematical physics.

Worked out problems

1. Express $f(x) = 4x^3 + 6x^2 + 7x + 2$ in terms of Legendre polynomials.

Solution: Let $f(x) = 4x^3 + 6x^2 + 7x + 2 = aP_3(x) + bP_2(x) + cP_1(x) + dP_0(x)$

We have the Rodrigue's formula for Legendre polynomials

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

Therefore, we get $P_0(x) = 1$, $P_1(x) = x$, $P_2(x) = \frac{1}{2}(3x^2 - 1)$, $P_3(x) = \frac{1}{2}(5x^3 - 3x)$

$$4x^3 + 6x^2 + 7x + 2 = a \frac{1}{2}(5x^3 - 3x) + b \frac{1}{2}(3x^2 - 1) + cx + d$$

Equating corresponding coefficients, we get

$$4 = \frac{5a}{2}, 6 = \frac{3b}{2}, 7 = -\frac{3a}{2} + c, 2 = -\frac{b}{2} + d$$

$$\Rightarrow a = \frac{8}{5}, b = 4, c = \frac{47}{5}, d = 4$$

Thus, $4x^3 + 6x^2 + 7x + 2 = \frac{8}{5}P_3(x) + 4P_2(x) + \frac{47}{5}P_1(x) + 4P_0(x)$.

2. Prove that $\int_{-1}^1 xP_n(x)P_{n-1}(x)dx = \frac{2n}{4n^2 - 1}$.

Solution: We have the recurrence relation

$$(2n-1)xP_{n-1}(x) = nP_n(x) + (n-1)P_{n-2}(x)$$

Multiplying by $P_n(x)$ and then taking limit of integration from -1 to $+1$, we get

$$(2n-1) \int_{-1}^1 xP_n(x)P_{n-1}(x)dx = n \int_{-1}^1 P_n^2(x)dx + (n-1) \int_{-1}^1 P_n(x)P_{n-2}(x)dx$$

By orthogonal property, we then get

$$(2n-1) \int_{-1}^1 xP_n(x)P_{n-1}(x)dx = n \times \frac{2}{2n+1} + (n-1) \times 0$$

$$\Rightarrow \int_{-1}^1 xP_n(x)P_{n-1}(x)dx = \frac{2n}{4n^2 - 1}$$

3. Prove that $\int_{-1}^1 xP_n(x)P_n'(x)dx = \frac{2n}{2n+1}$.

Solution: We have the recurrence relation

$$xP_n'(x) = nP_n(x) + P_{n-1}'(x) \quad (1)$$

$$P_{n+1}'(x) = (2n+1)P_n(x) + P_{n-1}'(x) \quad (2)$$

Replacing n by $n-2$, $n-4, \dots$ in (2), we get

$$P_{n-1}'(x) = (2n-3)P_{n-2}(x) + P_{n-3}'(x) \quad (3)$$

$$P_{n-3}'(x) = (2n-7)P_{n-4}(x) + P_{n-5}'(x) \quad (4)$$

and so on.

Adding (1), (3) and (4), we get

$$xP_n'(x) = nP_n(x) + (2n-3)P_{n-2}(x) + (2n-7)P_{n-4}(x) + \dots$$

Multiplying by $P_n(x)$ and then taking limit of integration from -1 to $+1$, we get

$$\begin{aligned} \int_{-1}^1 xP_n(x)P_n'(x)dx &= n \int_{-1}^1 P_n^2(x)dx + (2n-3) \int_{-1}^1 P_n(x)P_{n-2}(x)dx + (2n-7) \int_{-1}^1 P_n(x)P_{n-4}(x)dx + \dots \\ \int_{-1}^1 xP_n(x)P_n'(x)dx &= n \times \frac{2}{2n+1} + (2n-3) \times 0 + (2n-7) \times 0 + 0 \\ \Rightarrow \int_{-1}^1 xP_n(x)P_n'(x)dx &= \frac{2n}{2n+1}. \end{aligned}$$

$$4. \text{ Prove that } \int_{-1}^1 x^2 P_n^2(x)dx = \frac{1}{8(2n-1)} + \frac{3}{4(2n+1)} + \frac{1}{8(2n+3)}.$$

Solution: We have the recurrence relation

$$(2n-1)xP_{n-1}(x) = nP_n(x) + (n-1)P_{n-2}(x) \quad (1)$$

Replacing n by $n+1$, we get

$$(2n+1)xP_n(x) = (n+1)P_{n+1}(x) + nP_{n-1}(x) \quad (2)$$

Squaring both sides and then taking limit of integration from -1 to $+1$, we get

$$\begin{aligned} (2n+1)^2 \int_{-1}^1 x^2 P_n^2(x)dx &= (n+1)^2 \int_{-1}^1 P_{n+1}^2(x)dx + n^2 \int_{-1}^1 P_{n-1}^2(x)dx + 2n(n+1) \int_{-1}^1 P_{n+1}(x)P_{n-1}(x)dx \\ &= (n+1)^2 \times \frac{2}{2(n+1)+1} + n^2 \times \frac{2}{2(n-1)+1} + 2n(n+1) \times 0, \text{ by orthogonal property} \end{aligned}$$

$$\begin{aligned} \text{Therefore, } \int_{-1}^1 x^2 P_n^2(x)dx &= \frac{1}{(2n+1)^2} \left[\frac{2(n+1)^2}{2n+3} + \frac{2n^2}{2n-1} \right] \\ &= \frac{1}{(2n+1)^2} \times \frac{2(n^2+2n+1)(2n-1) + 2n^2(2n+3)}{(2n+3)(2n-1)} \\ &= \frac{1}{(2n+1)^2} \times \frac{8n^3+12n^2-2}{(2n+3)(2n-1)} = \frac{(2n+1)(4n^2+4n-2)}{(2n+1)^2(2n-1)(2n+3)} \\ &= \frac{4n^2+4n-2}{(2n+1)(2n-1)(2n+3)} \\ &= \frac{1}{8(2n-1)} + \frac{3}{4(2n+1)} + \frac{1}{8(2n+3)}. \end{aligned}$$

Problems on Legendre Functions

1. Prove that $(1-2xh+h^2)^{-\frac{1}{2}}$ is the generating function for Legendre's polynomials.

2. Use the generating function of Legendre polynomials to prove

$$(i) \quad nP_n(x) = (2n-1)xP_{n-1}(x) - (n-1)P_{n-2}(x)$$

$$(ii) \quad xP_n'(x) - P_{n-1}'(x) = nP_n(x)$$

$$(iii) \quad P_n'(x) - xP_{n-1}'(x) = nP_{n-1}(x)$$

$$(iv) \quad P_{n+1}'(x) - P_{n-1}'(x) = (2n+1)P_n(x)$$

$$(v) \quad (1-x^2)P_n'(x) = n(P_{n-1}(x) - xP_n(x))$$

$$(vi) \quad (x^2-1)P_n'(x) = (n+1)[P_{n+1}(x) - xP_n(x)].$$

3. Prove that $\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1}$ if $m = n$.

4. Prove that $\int_{-1}^1 x^2 \{P_n(x)\}^2 dx = \frac{1}{8(2n-1)} + \frac{3}{4(2n+1)} + \frac{1}{8(2n+3)}$.

5. Show that $\int_{-1}^1 x^2 P_{n-1}(x) P_{n+1}(x) dx = \frac{2n(n+1)}{(4n^2-1)(2n+3)}$.

6. Show that $\int_{-1}^1 (x^2-1)P_{n+1}(x) P_n'(x) dx = \frac{2n(n+1)}{(2n+1)(2n+3)}$.

7. Show that $\int_{-1}^1 xP_n(x) P_{n-1}(x) dx = \frac{2n}{4n^2-1}$.

8. Show that $\int_{-1}^1 (1-x^2)[P_n'(x)]^2 dx = \frac{2n(n+1)}{2n+1}$.

9. Show that $\int_{-1}^1 xP_n(x) P_n'(x) dx = \frac{2n}{2n+1}$.

10. State and prove the Rodrigue's formula.

11. Write the Legendre polynomials of $P_n(x)$ for $n = 0, 1, 2, 3, 4, 5$.

$$\text{Even polynomials: } P_0(x) = 1, P_2(x) = \frac{1}{2}(3x^2 - 1), P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3)$$

$$\text{Odd polynomials: } P_1(x) = x, P_3(x) = \frac{1}{2}(5x^3 - 3x), P_5(x) = \frac{1}{8}(63x^5 - 70x^3 + 15x)$$

12. Express $f(x) = x^4 + 2x^3 + 2x^2 - x - 3$ in a series of Legendre polynomials.

13. Express $f(x) = 4x^3 + 6x^2 + 7x + 2$ in terms of Legendre polynomials.

Problems on Bessel Functions

1. Prove that $J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$ and $J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x$.

2. Use recurrence formula $2n J_n(x) = x[J_{n-1}(x) + J_{n+1}(x)]$ to prove that

$$(i) \quad J_{-\frac{3}{2}}(x) = -\sqrt{\frac{2}{\pi x}} \left(\frac{\cos x}{x} + \sin x \right)$$

$$(ii) \quad J_{\frac{3}{2}}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x \right)$$

3. Prove that $x J_n'(x) = n J_n(x) - x J_{n+1}(x) = -n J_n(x) + x J_{n-1}(x)$.

4. Prove that $2 J_n'(x) = J_{n-1}(x) - J_{n+1}(x)$.

5. Prove that $2n J_n(x) = x[J_{n-1}(x) + J_{n+1}(x)]$.

6. Prove that $\frac{d}{dx}[x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$.

7. Prove that $\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$.

8. Prove that $\frac{d}{dx}[J_2(x)] = \left(1 - \frac{4}{x^2}\right) J_1(x) + \frac{2}{x} J_0(x)$, where $J_n(x)$ is the Bessel's function of order n .

9. Prove that $\frac{d}{dx}[x J_n(x) J_{n+1}(x)] = x[J_n^2(x) - J_{n+1}^2(x)]$.

10. Prove that $\frac{d}{dx}[J_n^2(x) + J_{n+1}^2(x)] = 2 \left[\frac{n}{x} J_n^2(x) - \frac{n+1}{x} J_{n+1}^2(x) \right]$.

11. Prove that $\frac{d}{dx}[J_n(x)] = J_{n-1}(x) - \frac{n}{x} J_n(x) = \frac{n}{x} J_n(x) - J_{n+1}(x)$ and hence prove that

$$J_n(x) = \frac{x}{2n} [J_{n-1}(x) + J_{n+1}(x)].$$

12. Prove that $\int_0^x x^2 J_0(x) J_1(x) dx = \frac{x^2}{2} J_1^2(x)$.

13. Show that $\int_0^1 x^3 J_0(\alpha x) dx = \frac{\alpha^2 - 4}{\alpha^3} J_1(\alpha) + \frac{2}{\alpha^2} J_0(\alpha)$.

14. Prove the orthogonal property $\int_0^1 x J_n(\alpha x) J_n(\beta x) dx = \begin{cases} 0 & \text{if } \alpha \neq \beta \\ \frac{1}{2} [J_{n+1}(\alpha)]^2 & \text{if } \alpha = \beta \end{cases}$

where α and β are the roots of $J_n(x) = 0$.